

JENNIFER DUONG

Software Engineer with hands-on experience in full-stack web development, high-performance computing, and applied ML, building scalable systems across cloud, research, and product environments.

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PROFESSIONAL EXPERIENCE

InvestorHub

Melbourne CBD, VIC

Junior Software Engineer

Dec. 2025 – Present

- Built an in-house video recording tool using Chrome's native MediaRecorder API, enabling companies to capture and embed video content directly within the platform.
- Refactored a content block system to consolidate duplicated code across style variants, improving maintainability and making it significantly easier to extend with new variants in the future.
- Contributed full-stack features across the marketing suite: newsflow distribution (email, LinkedIn, Twitter), a flexible website builder, and webinar tooling integrated with shareholder data.

University of Melbourne & CSL

Carlton, VIC

Technical Research Assistant

Aug. 2024 - Nov. 2025

- Developed parallelized machine learning scripts in Python, executed on University of Melbourne's HPC Spartan to accelerate research workflows.
- Built FastAPI applications to operationalize computational models developed by PhD researchers, integrating data in and out via Databricks, hosted on Posit Connect for easy access.

InvestorHub

Melbourne CBD, VIC

Software Engineer Intern

Nov. 2023 - Feb. 2024

- Built a demo website from scratch emulating InvestorHub services, featuring AI-generated questions and answers, now used in sales meetings to showcase the platform (React frontend, Elixir backend, PostgreSQL on GCP).
- Developed a monthly dashboard in React with Tailwind CSS, enabling companies to download summaries of InvestorHub's impact on their reach.
- Created a statistics page displaying backend data as graphs to visualize growth in shareholders and user engagement.

AllMediaDesk

Melbourne CBD, VIC

Software Engineer Intern

Feb. 2023 - Sept. 2023

- Partnered with sales and release teams to fix performance and usability issues, develop a CRM portal (TSX frontend, JavaScript backend), and containerize AdPerform and AdOptimize, enhancing customer engagement and reducing upgrade times.

EDUCATION

University of Melbourne

Carlton, VIC

Masters of Computer Science, Distributed Computing – First Class Honors (83%)

Feb. 2024 – Dec. 2025

Relevant subjects: Cluster and Cloud Computing, AI Planning for Autonomy, Distributed Systems, Mobile Computing

University of Melbourne

Carlton, VIC

Bachelor of Science, Computer and Software Systems – Second Class Honors (79%)

Feb. 2021 – Dec. 2023

TECHNICAL PROFICIENCY

Languages & Frameworks: Python, TypeScript (TSX), JavaScript, Elixir, Java, React, Next.js, Node.js, HTML/CSS, GraphQL

Cloud & Infrastructure: FastAPI, PostgreSQL, MySQL, Databricks, GCP, Docker, Posit Connect, HPC (Spartan/SLURM), Git, Tailwind CSS

PUBLICATIONS

Duong QCT, Golzarjalal M, Aickelin U, Otte E. Jun. 2026

A Generalized Framework for Multi-fidelity Acquisition Functions for Bioprocessing Applications. *Proceedings of the 17th ACM International Conference on Bioinformatics, Computational Biology and Health Informatics (ACM-BCB '26)*, 1–10.
doi:10.1145/3807503.3819466

Golzarjalal M, Aickelin U, **Duong QCT**, Otte E. Feb. 2026

Doing More with Less: Multifidelity Optimization in the Biopharmaceutical Industry. *BioProcess International*, 24(2).

Golzarjalal M, et al. (incl. **Duong QCT**) Aug. 2025

A Dynamic and Generalizable Modelling Framework Based on Genome-Scale Flux Balance Analysis for CHO Fed-Batch Culture. *SSRN preprint*. doi:10.2139/ssrn.5591701

VOLUNTEER & EXTRACURRICULAR

Student Ambassador — University of Melbourne Mar. 2025 – Present

- Represented the Faculty at Open Day and other outreach events; developed leadership and public-speaking skills through panels, tours, and content creation.

Melbourne Plus: People Leadership — University of Melbourne Aug. 2025

REFERENCES

- Jimmy Qiu, InvestorHub Head of Engineering
- Mohammad Golzarjalal, University of Melbourne
- Akil Pitchandi, AllMediaDesk Software Developer and Project Manager
- Contact details available on request.